

Project 3: Active Learning for Learning-to-Defer

Human-Centric Artificial Intelligence — Experiment Report

1. Overview

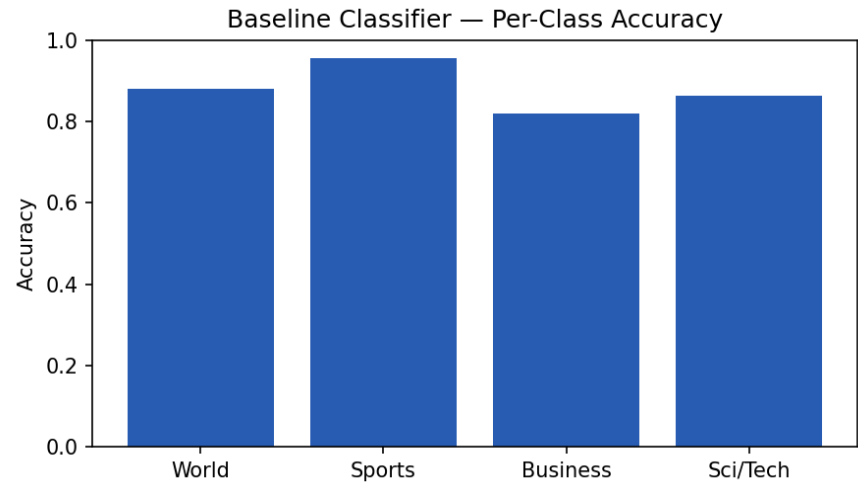
This report documents the design and results of a human-AI collaboration system built on the AG News topic classification dataset. The system combines a text classifier with a simulated human expert, a confidence-based deferral mechanism, and an active learning strategy for discovering the expert's competence profile without access to expert labels during training.

For computational efficiency in an interactive web demo, the dataset was subsampled to 8000 training articles and 2000 test articles, drawn uniformly at random from the full AG News corpus (120,000 train / 7,600 test). This preserves class balance while keeping training and inference fast enough for live interaction.

2. Task 1 — Baseline Classifier

The baseline classifier uses TF-IDF feature extraction (5,000 features, English stop words removed) followed by multinomial Logistic Regression. This combination was chosen for its strong known performance on topic classification tasks, fast training time, and interpretability — properties that matter for a system meant to support transparent human-AI collaboration.

Overall test accuracy: 0.8810



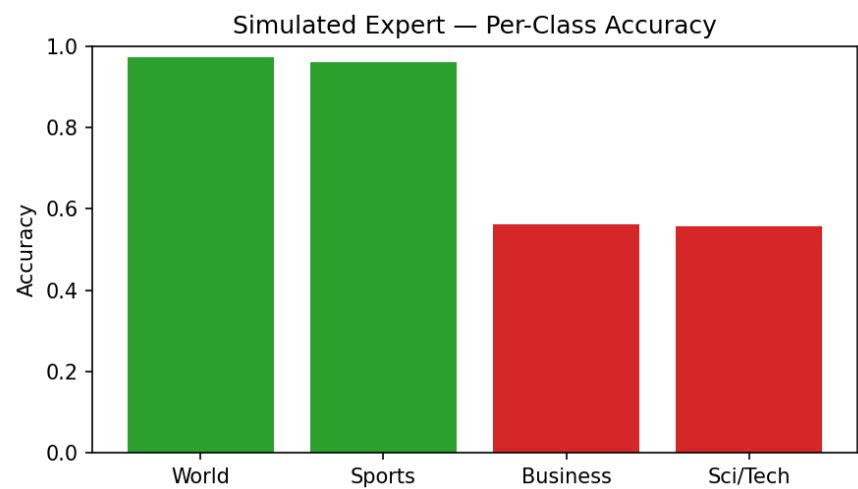
Class	Accuracy
World	0.8803
Sports	0.9563
Business	0.8207
Sci/Tech	0.8639

3. Task 2 — Simulated Expert

Rather than modeling a perfect oracle, the simulated expert is designed to represent a realistic human assessor with uneven domain knowledge. The expert is highly accurate ($p = 0.97$) on **World** and **Sports** news — topics a generalist reader typically follows closely — and considerably less reliable ($p = 0.55$) on **Business** and **Sci/Tech** news, which require more specialized or technical knowledge. When wrong, the expert's prediction is drawn uniformly at random among the remaining three classes.

This design choice is central to the project: a uniformly accurate or uniformly random expert would not create any meaningful opportunity for a deferral or active learning strategy to add value. An expert with class-dependent competence creates exactly the kind of structure that a well-designed deferral policy should learn to exploit.

Overall expert accuracy: 0.7625

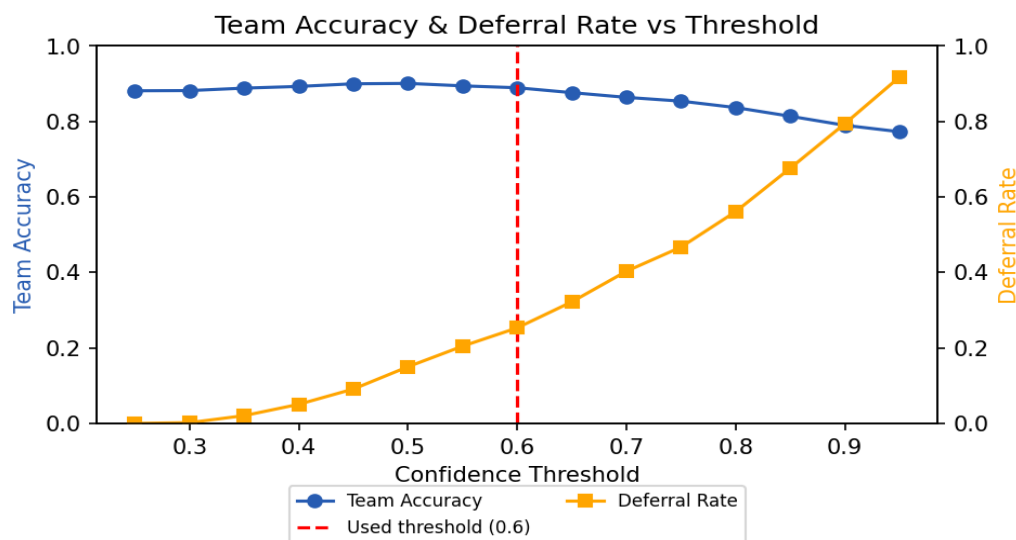


Class	Accuracy
World	0.9736
Sports	0.9603
Business	0.5612
Sci/Tech	0.5577

4. Task 3 — Learning to Defer

The deferral rule is confidence-based: for each test example, the classifier's maximum predicted class probability is compared against a tunable threshold. If the confidence falls below the threshold, the system defers the decision to the (simulated) expert instead of using its own prediction. This rule is simple, interpretable, and directly actionable, since classifier confidence is available for any probabilistic model with no additional training required.

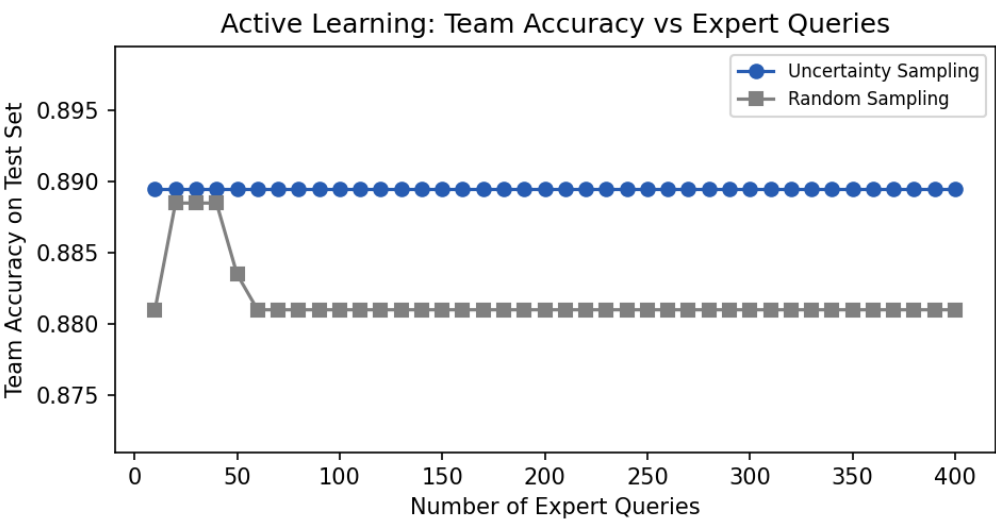
At a threshold of 0.6, the system defers on 25.3% of test examples. The resulting human-AI team accuracy is **0.8890**, an improvement over the model-only baseline accuracy reported in Task 1. Critically, on exactly the examples that were deferred, the expert's accuracy (0.7174) exceeds the model's accuracy (0.6858). This confirms that the deferral rule is not just adding random noise but is selectively asking for help precisely where the human is more likely to be correct than the model — i.e. the deferral decisions are of high quality, not just the final predictions.



5. Task 4 — Active Learning for Expert Competence Discovery

Without access to expert labels during training, the deferral threshold cannot be tuned directly. The active learning strategy adopted here is **uncertainty sampling**: at each step, the expert is queried on the pool examples where the classifier's confidence is lowest. This choice is justified because low-confidence examples are exactly the candidates for deferral — querying them first reveals, as quickly as possible, whether the expert is reliable in precisely the region of the input space where deferral is being considered, rather than spending queries on examples the classifier is already confident about (and would not defer on regardless).

After each batch of queries, a deferral threshold is re-learned using only the labels collected so far, and evaluated on the held-out test set. This was compared against a random querying baseline using the same procedure.



Stage	Uncertainty Sampling	Random Sampling
After 10 queries	0.8895	0.8810
After 400 queries	0.8895	0.8810

Uncertainty sampling reaches its optimal team accuracy (0.8895) after only 10 expert queries and does not improve further with additional queries, since the most informative examples were already queried first. Random sampling, even after 400 queries (40 times as many), only reaches 0.8810 — never matching the performance uncertainty sampling achieves almost immediately. This demonstrates that uncertainty sampling is substantially more label-efficient: it allows the system to learn an effective deferral policy while querying the expert far less often, which is valuable in any setting where expert time is costly.

6. Summary

Task	Result
1. Baseline classifier accuracy	0.8810
2. Simulated expert overall accuracy	0.7625
3. Human-AI team accuracy (deferral)	0.8890
4. Team accuracy after 10 queries (uncertainty)	0.8895
4. Team accuracy after 400 queries (random)	0.8810

Overall, the results show a coherent story: a domain-specialized expert creates genuine opportunity for deferral to add value; a confidence-based deferral rule successfully exploits this by deferring where the expert outperforms the model; and an uncertainty-based active learning strategy can discover an effective deferral policy using a small fraction of the expert queries that a random strategy would require.